

Building the Perfect City: A Mathematical Blueprint

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How Geometry, Graph Theory, and Optimisation Can Design the Ideal Urban Space.

What makes a perfect city? Should it be a grid like New York, a radial hub like Paris, or something entirely different? Cities naturally evolve over time, often leading to inefficiency, high infrastructure costs, and excessive energy consumption, but mathematics offers a way to design them systematically — optimising infrastructure, transport, and public services.

Suppose we have the opportunity to bulldoze a city of our choosing, perhaps we'd choose Slough or Hull. We'll also assume we have an incredibly large but finite budget, the leftovers of which we could keep (this gives us the incentive to be efficient and economical whilst planning our city, whilst still allowing us to explore the possibilities to their fullest extent). We now have the opportunity to create our own "perfect" city, on the ruins of the unlucky place we just bulldozed, using mathematical principles.

In this article we will explore how geometry, graph theory, and optimisation techniques can help design an ideal city, ensuring efficient transport, minimal congestion, and maximum accessibility.

How to Decide a City Layout: Geometry

Before we start even building the city the first thing we'll have to decide on is the structure of the streets. The way a city is laid out affects many things including travel distances, infrastructure costs and land efficiency.

The three most common layouts seen around the world are:

Square Grid Layout — (New York, Milton Keynes)

A square grid is made of streets running parallel and perpendicular to each other, dividing the city into square blocks.

Some advantages to this layout are it is easy to navigate since streets run in a predictable way, and lend themselves to a numbered street naming system, it is also simple to build and expand.

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[Portland, Oregon](#)

However, travel distances are longer than necessary because movement is restricted to horizontal/vertical roads. If we imagine our square grid city as a grid on a graph with an x and y-axis then the distance between two points (A and B) in a square grid follows the Manhattan Distance formula:

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This is always longer than (or the same as) the true straight line (Euclidean) distance, making routes inefficient.

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[Background image is of Athens](#)

Radial Layout — (Paris, Washington D.C.)

A radial city has a central hub with roads spreading outward in a circular pattern, similar to spokes on a bicycle wheel. The main advantage is there is fast access to the city centre from any point. However, this layout causes congestion at central points and is difficult to expand without breaking the pattern.

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[Plaza Del Ejecutivo](#)

Organic Growth — (London, Tokyo)

Unlike planned cities, organically grown cities evolve naturally over centuries based on historical trade routes, geography, and social factors. These cities lack strict patterns and instead follow natural topography and historical growth patterns.

Advantages are a more vibrant urban environment made up of human scaled streets, this layout also adapts to natural landscape features. However, it can be chaotic, inefficient and hard to navigate due to irregular street patterns.

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[The City of London from above at night — Vincent LaForet](#)

The Case for Hexagons

Now this might be surprising, whilst hexagons may not seem like the most efficient shape for a city, they actually have

- The shortest perimeter for the same area (saving road materials)
- More direct travel paths (when compared to squares)
- Efficient use of space (seen in beehive and cell networks)

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Chat GPT's very best attempt at a city with hexagonal blocks

Assume we have two cities made of blocks of identical areas (A), but the first city has square blocks and the second city has hexagons.

For a square grid:

A square block of area A has side length

Each block has 4 sides, so therefore 4 roads, but each road is shared by 2 blocks so the street length per block is therefore

For a hexagonal grid:

A regular hexagon with area A has side length

Each hexagonal block has 6 sides, but each road is shared between 2 blocks, so the effective street length per block is

Substituting h in gives

Approximating

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Now we can compare the total road length per block

Cancelling A gives us

So the hexagonal grid city requires 93% of the street length of the square grid per block, meaning a 7% reduction. So a hexagonal city would be more efficient in terms of road infrastructure than a square grid city.

*If you're interested in hexagonal cities (and why the theoretical efficiency may not translate for practical purposes), [Hexagonal Planning in Theory and Practice](#) by Eran Ben-Joseph and David Gordon provides a more detailed analysis. I have also linked [this chapter](#) from *Fractal Cities* by Michael Batty and Paul Longley that also covers hexagonal cities.*

Where to Place Essential Services: Voronoi Diagrams

We will now explore how to place hospitals, schools and parks in a way that ensures fair access and even coverage. To achieve this we will use Voronoi diagrams, which can divide a city into regions where each point is closest to a specific facility.

How Voronoi diagrams work:

Given a set of points, the diagram partitions the space so that each region contains all locations closer to one point than any other.

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A GIF showing the creation of a Voronoi diagram, generated using the code from [this community post](#).

Using the Voronoi diagrams we can ensure that the hospitals are equally distributed, and we don't have one hospital support an over proportionate area. We can also ensure things such as no house being more than 15 minutes away from the nearest hospital.

Below we have an example of how this could be used, taken from a [paper on Auto-Differentiation of the Voronoi Tessellation](#).

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We can also use this technique to set school catchment areas and identify areas that need another school. Another use is also to maximise green space availability while ensuring equal coverage.

Urban Transport Optimisation: Graph Theory

Now we need to consider how people would move around our city, to do this we're going to use graph theory.

When you hear the word "graph", you might think of a line chart with an x and y-axis, but in this section we're looking at a different type of graph. In graph theory, a graph is a collection of points (called nodes or vertices) connected by lines (called edges).

We can therefore imagine a city's road and transport networks as a graph — a system of nodes (intersections or stations) and edges (roads or transit routes). Using graph theory, we can create more efficient transport systems and optimise traffic light timing for our city.

Public Transport Optimisation

A transport network can be represented as a weighted graph, where edge weights correspond with travel times or congestion levels, stations are nodes and tracks or bus routes between stations are edges.

Once we have this structure, mathematicians can analyse the efficiency of such a transport system; using [Dijkstra's Algorithm](#) to find shortest routes, using [graph centrality](#) to identify the most important stations, and exploring weak points of the system (if one station or track is closed, how badly does it affect the rest of the network?).

[Studies of the London Underground](#) found that certain stations are critical to the entire network. If one of these key nodes fail, large parts of the city become inaccessible. Graph theory will help us avoid issues like this when building our city.

Traffic Light Synchronisation

Traffic lights at intersections can also form a graph, each intersection is a node and each road connecting two intersections is an edge. The goal is to assign different “colours” (or timing patterns) to each intersection so that traffic flows smoothly.

This is a graph colouring problem, a mathematical technique where adjacent intersections (nodes) must have different timings (colours) to prevent congestion. By using graph theory, cities like New York and Singapore have designed smart traffic light systems that adjust in real time, preventing gridlock.

Energy Efficiency: Sun Path Optimisation

The arrangement of buildings determines how much sunlight they receive, which affects solar panel efficiency, energy costs and heat absorption (reducing cooling costs in summer)

Optimal Building Orientation

For maximum solar energy, buildings should:

1. Face the equator (south facing in the Northern Hemisphere, north facing in the Southern Hemisphere)
2. Use angled rooftops, optimised for seasonal sunlight changes
3. Avoid self shading by staggered building heights

Mathematical Models for Solar Efficiency

Using computational models, we can simulate how sunlight interacts with buildings in our city. The [Ray tracing algorithms](#) predict how shadows move throughout the day, best location for solar panels, and ideal building spacing to avoid shading.

One of the most important factors in solar energy generation is the angle at which solar panels are installed. The energy received by a solar panel is calculated by the following formula:

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Where:

- E is the energy collected
- A is the solar panel area
- S is the [solar irradiance](#) (power per unit area received from the Sun measured in W/m²)
- θ is the angle between the panel and the sun's rays

We know that the maximum value of $\cos(\theta)$ is 1, and this occurs when $\theta = 0$ degrees. This means solar panels should be perpendicular to the incoming sunlight to capture the maximum possible energy. However, this doesn't mean they should be laying flat; the angle at which the sun will hit the panel also depends on the time of day, time of year and the latitude of the city. If we figured out where we want to build our city we could then use an online calculator like [this one](#) to figure out our optimal solar panel tilt.

When Perfection Fails: The Human Factor

Our mathematical models discussed above can optimise nearly every aspect of urban planning — from transport efficiency to solar energy usage. But cities are not just mathematical constructs; they are places for people. No matter how mathematically “perfect” a city appears on paper, human behaviour, culture and history shape how it functions in reality.

While our city designed purely on mathematical principals may seem optimal, we should consider the below examples to see when rigidly planned cities have failed to account for how people actually live.

Brasilia, Brazil

Designed with a symmetrical layout, Brasilia was intended to be a model capital. However, its design prioritised cars over pedestrians, leading to transport inefficiencies and social divisions.

[The city's reliance on highways and broad streets over pedestrian pathways has been criticised for limiting spontaneous social interactions and making it difficult for low income residents to access central areas.](#)

[Over time, Brasilia has experienced urban sprawl and class inequality, with many residents living in lower income satellite cities far from the central district.](#)

Milton Keynes, UK

Built on a strict grid system, Milton Keynes boasts excellent road infrastructure but faces challenges with public transport and pedestrian accessibility.

[The city's grid system was designed for car use, resulting in high levels of car dependency and lower usage of public transport, contributing to unsustainable greenhouse gas emissions.](#)

[While Milton Keynes has an extensive network of 'redways' \(off-road cycleways and pedestrian footpaths\) these are underutilised, and pedestrian movement is often hampered by the engineered road infrastructure.](#)

Organically Grown Cities

Cities like London and Tokyo, which have developed organically over time, often appear chaotic but are more walkable and adaptable to human movement and behaviour.

Maybe we shouldn't bulldoze a city to create a mathematically perfect one just yet, we may have overlooked the most important factor: making it a place where people actually want to live.

A city is more than just roads, buildings, and transport systems. While efficiency is crucial, a truly great city also considers aesthetics, green spaces, and the overall quality of life. Not every design choice should be dictated by optimisation alone; some investments serve no mathematical purpose but make a city more vibrant, inviting, and liveable.

Maths and optimisation are powerful tools for designing better cities, but the goal isn't just efficiency — it's creating places where people thrive. A truly perfect city isn't just about numbers; it balances mathematical principles with human needs and adaptability.

This article is a piece of coursework for a university module so any feedback you could give on the Google Form below would be incredibly helpful. Thank you :)

Links in the article:

- [*Hexagonal Planning in Theory and Practice* by Eran Ben-Joseph and David Gordon](#)
- [*Fractal Cities* by Michael Batty and Paul Longley](#)
- [*Voronoi Diagram GIF code*](#)
- A Method for [*Auto-Differentiation of the Voronoi Tessellation*](#) by Sergei Shumilin, Alexander Ryabov, Evgeny Burnaev and Vladimir Vanovskii
- [*Dijkstra's Algorithm*'s Wikipedia page](#)
- [*Graph centrality* Wikipedia page](#)
- [*A network-based framework for assessing infrastructure resilience: a case study of the London metro system*](#) by Shauhrat S. Chopra, Trent Dillon, Melissa M. Bilec and Vikas Khanna.
- [*Ray tracing algorithm* Wikipedia page](#)
- [*Solar irradiance* Wikipedia page](#)
- [*Optimal Solar Panel Tilt Angle Calculator*](#)
- [*The Brazilianization of Brasilia* by Fernando Lara and Stella Nair](#)
- [*The Strike Zone: Brasilia — a modernist utopia?* by Eli Striker](#)
- [*SDG Sustainable Cities and Communities* — Milton Keynes Community Foundation](#)
- [*How the transport infrastructure of Milton Keynes affects the way people travel* by Georgina Marum and Alan Patterson](#)